When Does Causal Correction Help Discount Recommendation? Confounded Discounts in Negotiated Pricing

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Background: Sellers of negotiated products increasingly learn a win-probability model from past deals and use it to recommend discounts. The salespeople who set those past discounts knew things about each deal that the model does not, so discount is confounded with outcome. **Objectives:** We ask when this confounding changes the recommended discount, whether the common remedy of constraining the model to be monotone in discount repairs the decision, and when an instrument built from salesperson leniency helps or harms. **Methods:** We analyse a single-index model of rep-driven confounding and build a semi-synthetic benchmark with a known causal response, scored by revenue regret, with controls for the strength and direction of confounding, sample size, model capacity, and the validity of the instrument. **Results:** Mild confounding barely moves the recommendation; strong confounding collapses it, and which direction is harmful depends on the win model. A monotone constraint repairs the fitted curve but not the decision, and under a saturating response it lowers the recommendation and raises regret. A leniency control function is close to an oracle on large books when salespeople are assigned as if at random, and is worse than no correction when lenient salespeople differ in skill or receive systematically different deals. We prove that the first of these failures is undetectable from the deal data and show that standard balance and first-stage checks pass under the second. **Conclusions:** Causal correction for discount guidance is worth its variance only when assignment is rule-based and the book is large. A monotonicity belief is harmful as a constraint on an uncorrected model and useful as a falsification check on a corrected one.

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1 Introduction

A growing class of decision-support systems is trained on the logged decisions of the people it is meant to advise. A model observes what experts did and what happened, and then recommends what to do next. When the experts acted on information the model cannot see, a model that predicts outcomes well can still recommend badly, because the action and the outcome share a hidden cause.

Negotiated pricing is a clean instance. In business-to-business sales a representative (rep) chooses a discount for each deal. Vendors of pricing software fit a model of the probability of winning as a function of deal attributes and the discount, and recommend the discount that maximizes expected revenue. Reps discount with the deal in front of them in mind: a struggling deal may get a deeper discount to rescue it, or a strategically important one may get a deeper discount to make sure of it. Either way the discount carries information about the deal's prospects that is not recorded.

Practitioners know this. A common response is to constrain the win model to be non-decreasing in discount, on the argument that a larger discount can never hurt the chance of winning, so a fitted curve that slopes down must be an artefact of confounding. The constraint is cheap, it is supported by standard gradient-boosting libraries,

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and it produces curves that are easy to defend to a customer. This paper started from a simple question about that practice: does forcing the curve to have the right sign fix the recommendation?

It does not. We show three things.

(1) **The constraint repairs the picture, not the decision.** In a linear-index model the monotone-constrained and unconstrained models recommend the same discount for every strength and direction of confounding (Corollary 3.4). When the true response saturates, the constrained model recommends weakly less discount than the unconstrained one and has weakly higher regret whenever the unconstrained model is already under-discounting (Proposition 3.5). In our benchmark this costs 4 to 11 points of revenue under moderate to strong confounding at the capacity of a production model.

(2) **Whether confounding matters depends on the model as much as on the data.** Regret is flat in the bias for small biases and then rises sharply (Proposition 3.6). The direction of confounding that is harmful flips between a model that under-fits the response and one that fits it well. At realistic sample sizes, the capacity of the win model moves regret by more than confounding does.

(3) **An instrument from rep leniency works under conditions that sales organizations rarely guarantee, and its failures are silent.** A control function built on leave-one-out rep leniency is within a point of an oracle on books of 20,000 deals with randomly assigned reps. If lenient reps are also weaker or stronger sellers, the corrected model is worse than the uncorrected one, and we prove that no test on the deal data can detect it (Proposition 4.1). If managers route deals to reps on unrecorded deal quality, the same happens, and the balance test and first-stage diagnostics that the examiner-design literature recommends pass.

The contribution is an analysis and a benchmark, not a new estimator. The analysis says when correction changes the decision. The benchmark, which we release, lets methods for decision-focused learning and causal policy learning be tested under controlled violations of their assumptions, with regret measured against a known truth. Pricing is the worked setting; the questions arise wherever a shape constraint is used to make a model fitted to logged human decisions look causal.

2 Setting and Related Work

Customized pricing. In customized or negotiated pricing each customer receives an individual quote and accepts or declines (Phillips et al. 2015). The standard tool is a bid-response model, the probability of acceptance as a function of price. Phillips et al. (2015) study an indirect auto lender whose dealers negotiate rates, correct for price endogeneity with a control function (Petrin and Train 2010) using rates offered for similar loans in other regions as the instrument, and report that the correction raised their estimated price elasticities by 126% to 285%. Karlinsky-Shichor and Netzer (2024) model the pricing of individual salespeople at a business-to-business retailer, instrument price with product cost, and route quotes between model and salesperson. Neither uses the salesperson as the instrument, and neither studies shape constraints. Bockel-Rickermann et al. (2023) benchmark estimators of bid-response functions on semi-synthetic mortgage data under selection on observed covariates; our benchmark differs in having unobserved confounding, a rep structure, and regret in revenue terms. **[TODO: Read all three in full and check each characterization; add Miao et al. and Gupta and Kamble after reading.]**

Examiner designs. Using the leniency of a quasi-randomly assigned decision-maker as an instrument is standard in economics (Dobbie et al. 2018; Kling 2006). Frandsen et al. (2023) give a joint test of the exclusion and monotonicity assumptions, and Chan et al. (2022) show that such designs fail when decision-makers differ in skill and not only in threshold. Our Proposition 4.1 is the form that failure takes in pricing. **[TODO: Read Chyn, Frandsen and Leslie (2025) and cite.]**

Decisions versus predictions. Decision-focused learning trains models for the quality of the downstream decision (Elmachtoub and Grigas 2022; Mandi et al. 2024), assuming the training data sample the decision environment faithfully. Fernández-Loría and Provost (2022) argue that good decisions do not require accurate effect estimates and that confounded data can still support them. Our results agree for mild confounding and locate where the argument stops holding.

Robustness and deferral under hidden confounding. Kallus and Zhou (2021) learn policies that minimize worst-case regret over a sensitivity model, and Conley et al. (2012) relax the exclusion restriction to a bounded direct effect. Ghoummaid and Shalit (2024) learn when to defer to a human expert under bounded hidden confounding. We use a sensitivity range in the same spirit and find that choosing robustly dominates deferring to a default in our setting.

Shape constraints. Monotonicity is known to stabilize nonparametric instrumental-variable estimation (Chetverikov and Wilhelm 2017). We are not aware of an analysis of what a monotone constraint does to the decision when no instrument is used. **[TODO: Search again before submission.]**

3 A Model of Rep-Driven Confounding

We consider one cohort of deals; everything below holds conditional on observed covariates. A deal has latent quality $q \sim N(0,1)$, seen by the rep and not by the model. The rep sets the discount

$$d = m + \gamma q + \eta, \qquad \eta \sim N(0, \sigma^2) \text{ independent of } q, \tag{1}$$

and the deal is won if

$$Y = \mathbf{1}\{\alpha + g(d) + \beta q + \varepsilon > 0\}, \qquad \varepsilon \sim N(0,1), \tag{2}$$

with g increasing and $\beta > 0$. Write $\sigma_d^2 = \gamma^2 + \sigma^2$ and $\rho = \gamma/\sigma_d$ for the correlation between discount and quality. We call $\rho < 0$ *rescue* confounding (deeper discounts on weaker deals) and $\rho > 0$ *whale* confounding (deeper discounts on stronger deals). The causal win curve is

$$p^*(d) = \mathbb{P}(Y = 1 \mid do(d)) = \Phi\left(\frac{\alpha + g(d)}{\sqrt{1+\beta^2}}\right). \tag{3}$$

The seller chooses $d \in [d_{\text{lo}}, d_{\text{hi}}] \subset [0, 1)$ to maximize expected revenue per unit of list price, $R(d) = (1-d)\,p^*(d)$, with maximizer d^*. A model that believes a curve $\hat{p}$ recommends $\hat{d} = \arg\max_d (1-d)\hat{p}(d)$, taking the smallest maximizer, and its regret is $R(d^*) - R(\hat{d})$.

3.1 What a predictive model learns

Proposition 3.1 (Observational curve). *Under (1) and (2),*

$$\pi(d) := \mathbb{P}(Y = 1 \mid d) = \Phi\left(\frac{\alpha + g(d) + \beta\rho\,(d-m)/\sigma_d}{\sqrt{1 + \beta^2(1-\rho^2)}}\right). \tag{4}$$

If $g(d) = \theta d$ with $\theta > 0$, then π is a probit in d with slope $b_{\text{obs}} = (\theta + \beta\rho/\sigma_d)/\sqrt{1+\beta^2(1-\rho^2)}$, and: (i) $b_{\text{obs}} > b^* = \theta/\sqrt{1+\beta^2}$ whenever $\rho > 0$; (ii) $b_{\text{obs}} < 0$ if and only if $\rho < -\theta\sigma_d/\beta$; (iii) the confounding term vanishes at $d = m$, so the observational curve is the causal curve rotated about the mean historical discount, up to scale.

Proof. (q, d) is bivariate normal, so $q \mid d \sim N(\rho(d-m)/\sigma_d,\ 1-\rho^2)$. Write $q = \rho(d-m)/\sigma_d + \sqrt{1-\rho^2}\,\xi$ with $\xi \sim N(0,1)$ independent of d. Then $Y = 1$ if and only if $\alpha + g(d) + \beta\rho(d-m)/\sigma_d + \beta\sqrt{1-\rho^2}\,\xi + \varepsilon > 0$, and $\beta\sqrt{1-\rho^2}\,\xi + \varepsilon \sim N(0, 1 + \beta^2(1-\rho^2))$. Parts (i) to (iii) follow by inspection; for (i) both the numerator and the reciprocal of the denominator exceed their values at $\rho = 0$. □

Remark 1. Rescue confounding does not always flatten the curve on the index scale: the denominator in (4) shrinks as $|\rho|$ grows, and when $\theta\sigma_d \gg \beta$ this can outweigh the numerator. To first order in ρ the bias has the sign of ρ.

3.2 What a monotone constraint does

Let μ be the distribution of the historical discount on $[d_{\mathrm{lo}}, d_{\mathrm{hi}}]$, assumed to have a density that is positive throughout. A probabilistic classifier constrained to be non-decreasing in discount estimates, in the population, the function

$$\tilde{\pi} \in \arg\min_{f \text{ non-decreasing}} L(f), \qquad L(f) = -\int \{\pi \log f + (1-\pi)\log(1-f)\}\, d\mu, \tag{5}$$

over functions $f : [d_{\mathrm{lo}}, d_{\mathrm{hi}}] \to (0,1)$. We call $\tilde{\pi}$ the monotone fit. If π is itself non-decreasing then $\tilde{\pi} = \pi$, since π minimizes the integrand pointwise. The next lemma gives $\tilde{\pi}$ when π rises and then falls.

Lemma 3.2 (Monotone fit of a single-peaked curve). *Let $\pi : [d_{\mathrm{lo}}, d_{\mathrm{hi}}] \to (0,1)$ be continuous, non-decreasing on $[d_{\mathrm{lo}}, d_p]$ and non-increasing on $[d_p, d_{\mathrm{hi}}]$, with $d_p < d_{\mathrm{hi}}$. For $t \in [d_{\mathrm{lo}}, d_p]$ let $M(t) = \int_{(t,d_{\mathrm{hi}}]} \pi\, d\mu \,/\, \mu((t, d_{\mathrm{hi}}])$ be the mean of π to the right of t.*

(a) *If $\pi(d_{\mathrm{lo}}) \geq M(d_{\mathrm{lo}})$, then $\tilde{\pi}$ is the constant $\int \pi\, d\mu$.*
(b) *Otherwise there is $t \in (d_{\mathrm{lo}}, d_p]$ with $\pi(t) = M(t)$, and $\tilde{\pi} = \pi$ on $[d_{\mathrm{lo}}, t]$ and $\tilde{\pi} = \pi(t)$ on $(t, d_{\mathrm{hi}}]$.*

In both cases $\tilde{\pi}$ is unique up to μ-null sets.

Proof. *Existence of t in case (b).* The function $h(t) = \pi(t) - M(t)$ is continuous on $[d_{\mathrm{lo}}, d_p]$, because μ has a density and $\mu((t, d_{\mathrm{hi}}]) > 0$ there. By assumption $h(d_{\mathrm{lo}}) < 0$, and $h(d_p) \geq 0$ because $\pi(d_p)$ is the maximum of π. Hence $h(t) = 0$ for some $t \in (d_{\mathrm{lo}}, d_p]$.

The candidate. Let $A = [d_{\mathrm{lo}}, d_{\mathrm{hi}}]$ and $c = \int \pi\, d\mu$ in case (a), and $A = (t, d_{\mathrm{hi}}]$ and $c = \pi(t) = M(t)$ in case (b). Let $f^* = c$ on A and $f^* = \pi$ elsewhere. Then f^* is non-decreasing, takes values in a compact subset of $(0,1)$, and $\int_A (c - \pi)\, d\mu = 0$.

Sign structure. Let $s = \sup\{x \in A : \pi(x) \geq c\}$. We claim $\pi \geq c$ on $A \cap [d_{\mathrm{lo}}, s]$ and $\pi \leq c$ on $A \cap (s, d_{\mathrm{hi}}]$. The second holds by the definition of s. For the first, $\pi \geq c$ at the left end of A: in case (a) $c = \int \pi\, d\mu = M(d_{\mathrm{lo}})$, because μ has no atom at d_{lo}, and $\pi(d_{\mathrm{lo}}) \geq M(d_{\mathrm{lo}})$ by assumption; in case (b) $\pi(t) = c$. Moreover π is non-decreasing up to d_p and non-increasing on $[d_p, d_{\mathrm{hi}}]$, so the set where $\pi \geq c$ is an interval starting at the left end of A.

Optimality. For $\pi_0 \in (0,1)$ the function $\ell(u) = -\pi_0 \log u - (1-\pi_0)\log(1-u)$ is strictly convex on $(0,1)$ with $\ell'(u) = (u - \pi_0)/(u(1-u))$. So for any non-decreasing f and every x,

$$\ell_x(f(x)) \;\geq\; \ell_x(f^*(x)) + \frac{f^*(x) - \pi(x)}{f^*(x)(1 - f^*(x))}\,\big(f(x) - f^*(x)\big),$$

where ℓ_x is ℓ with $\pi_0 = \pi(x)$. The last term vanishes off A, where $f^* = \pi$. Integrating,

$$L(f) - L(f^*) \;\geq\; \frac{1}{c(1-c)} \int_A (c - \pi)(f - c)\, d\mu \;=\; \frac{1}{c(1-c)} \int_A (c - \pi)\, f\, d\mu,$$

using $\int_A (c - \pi)\, d\mu = 0$. For $x \in A$ with $x \leq s$ we have $c - \pi(x) \leq 0$ and $f(x) \leq f(s)$; for $x > s$ we have $c - \pi(x) \geq 0$ and $f(x) \geq f(s)$. In both cases $(c - \pi(x))f(x) \geq (c - \pi(x))f(s)$, so

$$\int_A (c - \pi)\, f\, d\mu \;\geq\; f(s) \int_A (c - \pi)\, d\mu = 0.$$

Hence $L(f) \geq L(f^*)$. Uniqueness follows from the strict convexity of ℓ. □

Proposition 3.3 (Monotone fit, linear index). *Let $g(d) = \theta d$. If $b_{\text{obs}} \geq 0$ then $\tilde{\pi} = \pi$. If $b_{\text{obs}} < 0$ then $\tilde{\pi}$ is the constant $\int \pi \, d\mu$, the overall win rate.*

Proof. If $b_{\text{obs}} \geq 0$ then π is non-decreasing. If $b_{\text{obs}} < 0$ then π is decreasing, so Lemma 3.2 applies with $d_p = d_{\text{lo}}$, and $\pi(d_{\text{lo}}) \geq M(d_{\text{lo}})$ puts us in case (a). □

Corollary 3.4 (Decision neutrality). *In the linear-index model the constrained and unconstrained models recommend the same discount for every ρ.*

Proof. If $b_{\text{obs}} \geq 0$ the two models coincide. If $b_{\text{obs}} < 0$, then under the unconstrained belief π is decreasing and under the constrained belief $\tilde{\pi}$ is constant; in both cases $(1-d)\hat{p}(d)$ is strictly decreasing and both recommend d_{lo}. □

So in the linear model the constraint changes what the user is shown, from a curve on which discounts hurt to one on which they do nothing, and leaves the recommendation and the regret unchanged. With a saturating response it does worse.

Proposition 3.5 (Saturating response). *Let g be increasing, concave and continuous on $[d_{\text{lo}}, d_{\text{hi}}]$, and write d_u and d_c for the recommendations of the unconstrained model, which believes π, and of the constrained model, which believes $\tilde{\pi}$.*

1. *If $\rho \geq 0$, π is increasing. If $\rho < 0$ it is single-peaked: non-decreasing up to some d_p and non-increasing afterwards. Where g is differentiable, π rises while $g'(d) > \beta|\rho|/\sigma_d$ and falls afterwards.*
2. *There is $t \in [d_{\text{lo}}, d_{\text{hi}}]$, with $t \leq d_p$ when $\rho < 0$, such that $\tilde{\pi} = \pi$ on $[d_{\text{lo}}, t]$ and $\tilde{\pi}$ is constant on $(t, d_{\text{hi}}]$.*
3. *$d_c \leq d_u$ and $d_c \leq t$. If $\rho \geq 0$ then $d_c = d_u$.*
4. *If $d_u \leq d^*$ then $R(d^*) - R(d_c) \geq R(d^*) - R(d_u)$, strictly if $d_c < d_u$.*

Proof. (1) By (4), $\pi(d) = \Phi(k(d)/s)$ with $k(d) = \alpha + g(d) + \beta\rho(d-m)/\sigma_d$ and $s > 0$. The function k is concave, being the sum of a concave and an affine function, and a concave function on an interval is non-decreasing up to a maximizer and non-increasing after it. Composition with the increasing map $\Phi(\cdot/s)$ preserves this. If $\rho \geq 0$, k is increasing. Where g is differentiable, $k'(d) = g'(d) + \beta\rho/\sigma_d$.

(2) If π is non-decreasing, $\tilde{\pi} = \pi$ and we take $t = d_{\text{hi}}$. Otherwise $d_p < d_{\text{hi}}$ and Lemma 3.2 applies, with $t = d_{\text{lo}}$ in its case (a).

(3) Write $W(d) = (1-d)\pi(d)$ and $\tilde{W}(d) = (1-d)\tilde{\pi}(d)$. If $\tilde{\pi} = \pi$ there is nothing to prove, and this covers $\rho \geq 0$. In case (a) of the lemma $\tilde{W}$ is strictly decreasing on $[d_{\text{lo}}, d_{\text{hi}}]$, so $d_c = d_{\text{lo}} \leq d_u$. In case (b), for $d > t$ we have $\tilde{W}(d) = (1-d)\pi(t) < (1-t)\pi(t) = \tilde{W}(t)$, so every maximizer of $\tilde{W}$ lies in $[d_{\text{lo}}, t]$, where $\tilde{W} = W$. Hence $d_c \leq t$, and d_c is the smallest maximizer of W over $[d_{\text{lo}}, t]$. If $d_u \leq t$, then d_u maximizes W over $[d_{\text{lo}}, t]$ and any other maximizer over $[d_{\text{lo}}, t]$ is a global maximizer of W, hence not smaller than d_u; so $d_c = d_u$. If $d_u > t$ then $d_c \leq t < d_u$.

(4) $\log R(d) = \log(1-d) + \log \Phi\big((\alpha + g(d))/\sqrt{1+\beta^2}\big)$. The first term is strictly concave. The second is concave, because $\log \Phi$ is concave and increasing and its argument is concave. So $\log R$ is strictly concave, d^* is unique, and R is strictly increasing on $[d_{\text{lo}}, d^*]$. If $d_c \leq d_u \leq d^*$ then $R(d_c) \leq R(d_u)$, with strict inequality when $d_c < d_u$. □

Part 4 has a condition. Lemma 3.7 below shows that mild rescue confounding can push d_u above d^*, and then a lower recommendation may help. Part 3 is specific to a single-peaked curve; with a dip between two high regions the constrained recommendation can be the higher one. Figure 1 illustrates the mechanism: the constrained model recommends the point at which its curve goes flat. The constraint converts "the data say discount stops helping here" into a hard stop, and under rescue confounding the data say so too early.

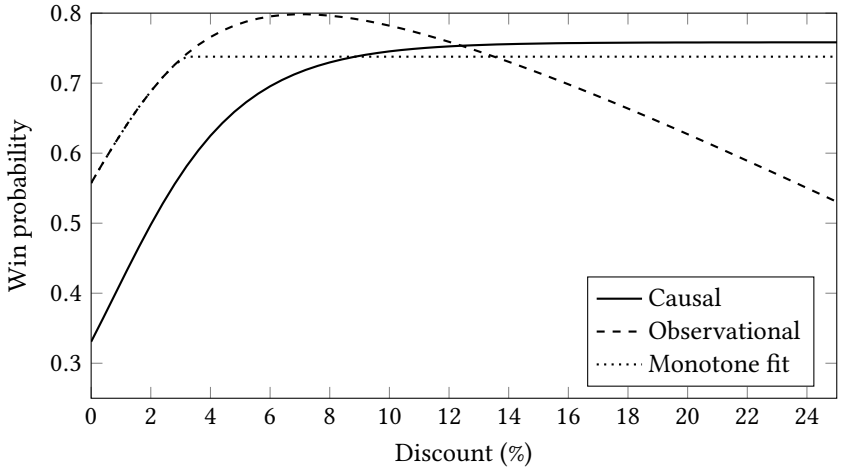


Fig. 1. Win curves under rescue confounding ($\rho = -0.5$, $\alpha = -0.5$, $\beta = 0.55$, mean historical discount 12%) with the saturating response used in the benchmark. The revenue-maximizing discount is 8.9% on the causal curve, 5.7% on the observational curve, and 3.1% on the monotone fit, which is where the fit goes flat.

3.3 When the bias changes the decision

Proposition 3.6 (Regret). *Let g be increasing and concave and let a model believe $\hat{p}(d) = \Phi(\hat{a} + \hat{b}d)$.*

(1) *Regret is at most* $\max\{R(d^*) - R(d_{\mathrm{lo}}),\ R(d^*) - R(d_{\mathrm{hi}})\}$.
(2) *If g is twice continuously differentiable and d^* is interior, regret equals* $\frac{1}{2}|R''(d^*)|(\hat{d} - d^*)^2 + o((\hat{d} - d^*)^2)$ *as* $\hat{d} \to d^*$.
(3) $\hat{d} = d_{\mathrm{lo}}$ *if and only if* $(1 - d_{\mathrm{lo}})\,\hat{b}\,r(\hat{a} + \hat{b}\,d_{\mathrm{lo}}) \le 1$, *where* $r = \phi/\Phi$. *For $\hat{a}$ in a bounded set this holds for all sufficiently small positive $\hat{b}$, so the recommendation reaches the floor before the fitted slope changes sign, and regret is then the constant* $R(d^*) - R(d_{\mathrm{lo}})$.

Proof. (1) By the proof of Proposition 3.5(4), $\log R$ is concave, so $R(\hat{d}) \ge \min\{R(d_{\mathrm{lo}}), R(d_{\mathrm{hi}})\}$ for every $\hat{d} \in [d_{\mathrm{lo}}, d_{\mathrm{hi}}]$.

(2) R is twice continuously differentiable and $R'(d^*) = 0$ at an interior maximum, so $R(\hat{d}) = R(d^*) + \frac{1}{2}R''(d^*)(\hat{d} - d^*)^2 + o((\hat{d} - d^*)^2)$ with $R''(d^*) \le 0$.

(3) Let $\hat{W}(d) = (1-d)\Phi(\hat{a} + \hat{b}d)$. Then $\log \hat{W}(d) = \log(1-d) + \log \Phi(\hat{a} + \hat{b}d)$ is strictly concave, with derivative $\psi(d) = -1/(1-d) + \hat{b}\,r(\hat{a} + \hat{b}d)$. A strictly concave function on $[d_{\mathrm{lo}}, d_{\mathrm{hi}}]$ is maximized at d_{lo} if and only if its derivative there is non-positive, and $\psi(d_{\mathrm{lo}}) \le 0$ is the stated condition. Since r is bounded on bounded sets, $\hat{b}\,r(\hat{a} + \hat{b}\,d_{\mathrm{lo}}) \to 0$ as $\hat{b} \downarrow 0$ uniformly over bounded $\hat{a}$, and $1/(1 - d_{\mathrm{lo}}) \ge 1$. □

Lemma 3.7 (Direction of the error). *Fix the value of the believed curve at m and vary its slope $\hat{b} > 0$. At an interior optimum with $z = \hat{a} + \hat{b}\hat{d}$,*

$$\operatorname{sign} \frac{\partial \hat{d}}{\partial \hat{b}} = \operatorname{sign}\big(1 - \hat{b}(\hat{d} - m)(z + r(z))\big).$$

Proof. Write $z = c + \hat{b}(d - m)$ with c fixed. With ψ as in the proof of Proposition 3.6, an interior optimum solves $F(d, \hat{b}) := (1 - d)\psi(d) = (1 - d)\,\hat{b}\,r(z) - 1 = 0$. Using $r'(z) = -r(z)(z + r(z))$,

$$\frac{\partial F}{\partial \hat{b}} = (1 - d)\,r(z)\big[1 - \hat{b}(d - m)(z + r(z))\big], \qquad \frac{\partial F}{\partial d} = -\psi(d) + (1 - d)\psi'(d) = (1 - d)\psi'(d) < 0$$

at the optimum, since $\psi = 0$ there and ψ is strictly decreasing. By the implicit function theorem $\partial \hat{d}/\partial \hat{b} = -(\partial F/\partial \hat{b})/(\partial F/\partial d)$, which has the sign of $\partial F/\partial \hat{b}$, and $(1 - d)r(z) > 0$. □

When the optimum is at or below the mean historical discount, a flatter curve lowers the recommendation and a steeper one raises it. When the optimum is well above the mean the directions reverse: a steeper curve makes the model believe the deal is already won just past the pivot. So "rescue confounding causes under-discounting" is not a general statement. Together with Proposition 3.6 this gives the practical picture: modest biases in the slope cost little, and the damage is concentrated where the recommendation is pushed to a boundary.

4 Correcting with Rep Leniency, and Why It Can Fail Silently

Suppose deals are handled by reps who differ in how much they discount for reasons unrelated to any one deal. Extend (1) and (2) to

$$d = f(x) + b_r + v, \qquad Y = \mathbf{1}\{h(x) + g(d) + \delta s_r + \beta q + \varepsilon > 0\}, \tag{6}$$

where b_r is the leniency and s_r the selling skill of rep r, x are covariates, and (v, q, ε) are independent of (x, r) but may depend on each other. If $\delta = 0$ (exclusion) and reps are assigned independently of (v, q, ε), leniency is a valid instrument. We estimate it by the leave-one-out mean, over a rep's other deals, of the discount residualized on covariates (Dobbie et al. 2018); the first-stage residual $\hat{v}$ enters the win model as a control function (Blundell and Powell 2004); and the causal curve is the fitted probability averaged over the distribution of $\hat{v}$.

Proposition 4.1 (Leniency-aligned skill is undetectable). *In model (6):*

(1) *If $s_r = \kappa b_r$, the same data are generated, deal by deal, by*

$$Y = \mathbf{1}\{\tilde{h}(x) + \tilde{g}(d) + \tilde{u} > 0\}, \quad \tilde{g}(d) = g(d) + \delta\kappa d, \quad \tilde{h}(x) = h(x) - \delta\kappa f(x), \quad \tilde{u} = \beta q - \delta\kappa v + \varepsilon,$$

in which there is no rep skill and $\tilde{u}$ is independent of (x, r).

(2) *The two models have different causal responses to discount, $g'(d)$ and $g'(d) + \delta\kappa$.*

(3) *If $s_r = s(b_r)$ for any function s, then Y is independent of r given (x, d, v).*

Proof. For (1), substitute $b_r = d - f(x) - v$ into $\delta\kappa b_r$ and collect terms. (2) is immediate. For (3), $b_r = d - f(x) - v$ is a function of (x, d, v), so conditioning on (x, d, v) fixes s_r, and the remaining randomness in Y is (q, ε) given v, which does not depend on r. □

The second model in part 1 satisfies every assumption of the instrument, so no test based on covariates, rep identity, discount and outcome can distinguish it from the first, while a control function fitted to the data recovers $\tilde{g}$. Part 3 says that tests for residual differences between reps have no power against skill that is a function of leniency. Writing $s_r = \kappa b_r + e_r$, only the component e_r unrelated to leniency is visible.

Remark 2 (Sorting). If managers route deals to reps on unrecorded quality, $\mathbb{E}[q \mid r]$ varies with b_r and enters the outcome equation exactly as δs_r does. Our experiments treat this case separately because it is the more plausible one in sales organizations.

Remark 3 (A use for monotonicity). When $\delta\kappa < 0$ (lenient reps are weaker) and g saturates, $\tilde{g}$ is eventually decreasing: the model without rep skill implies that large discounts lose deals. An analyst who believes discounts never hurt can reject it. So the monotonicity belief that is harmful as a constraint on the uncorrected model is informative as a falsification check on the corrected one. It gives no leverage when $\delta\kappa > 0$.

Table 1. Regret (% of optimal revenue; mean and standard error over 30 books) of a tree win model with and without a monotone constraint on discount, 20,000 training deals. "Difference" is constrained minus unconstrained, paired by book. "Clip" restricts recommendations to the segment's observed discount range.

ρ	Constrained	Unconstrained	Difference	Constrained (clip)	Difference (clip)
−0.9	30.5 (1.0)	19.5 (1.2)	+11.0 (1.0)	5.2 (0.3)	+0.9 (0.1)
−0.6	22.1 (0.8)	13.9 (0.8)	+8.2 (0.6)	3.9 (0.2)	+0.7 (0.1)
−0.3	15.9 (0.7)	11.7 (0.6)	+4.3 (0.5)	3.0 (0.2)	+0.3 (0.1)
0	9.3 (0.6)	8.7 (0.5)	+0.6 (0.3)	1.8 (0.1)	0.0 (0.0)
+0.3	4.2 (0.4)	4.3 (0.3)	−0.1 (0.3)	1.0 (0.1)	−0.1 (0.0)
+0.6	1.8 (0.2)	2.2 (0.2)	−0.4 (0.2)	0.6 (0.1)	−0.1 (0.1)
+0.9	1.4 (0.2)	2.1 (0.3)	−0.7 (0.2)	0.7 (0.1)	−0.4 (0.1)

A sensitivity range. Following Conley et al. (2012), we refit the control function assuming leniency has a direct effect δ on the index, for δ in a range $[-\Delta, \Delta]$, and recommend the discount that minimizes the maximum regret across the fitted models. The range is either fixed by the analyst or calibrated to the data by setting $\Delta = \hat{\tau}$, the estimated standard deviation of residual rep effects: the assumption is that hidden, leniency-aligned skill is no larger than the skill differences that are visible.

5 Benchmark

The benchmark generates books of closed deals with a known causal response. Each deal has a segment, region, industry, list price, competitor indicator and quarter-end indicator, drawn as in the synthetic generator of a commercial deal-analytics product; a rep; and a latent quality. The discount is the sum of a segment base level, competitor and quarter-end effects, the rep's bias, and a residual with standard deviation 3.5 points whose correlation with latent quality is the dial ρ. The win probability is logistic in a covariate index, the latent quality, and a saturating effect of discount, $1.3\tanh(d/0.05)$, which is essentially exhausted by a 10% discount. The spread of discounts is the same at every ρ, so only the reason for discounting changes.

Further dials are the weight of latent quality in the outcome, the number of training deals, the number of reps, a rep skill effect and its correlation with leniency, and the rule assigning reps to deals: at random, by segment (territories), or by a noisy reading of latent quality (sorting).

Because the response is known, the causal curve for each deal is computed by integrating over latent quality, and each recommendation is scored by regret: the revenue lost against the best discount for that deal, as a percentage of the revenue at that best discount, weighted by list price. We report regret for an unrestricted sweep over discounts from 0 to 40% and, where stated, for a sweep clipped to the 5th to 95th percentile of the discounts observed in the deal's segment, a guardrail used in practice. All results are means over 30 independently generated books with standard errors, except the capacity sweep (10 books). Recommendations are evaluated on fresh deals.

[TODO: State licence and repository. Add the real-covariate check on public auto-lending data if the dataset can be obtained.]

6 Experiments

6.1 The monotone constraint

We train gradient-boosted trees **[TODO: cite LightGBM]** with the configuration of a production win model (50 trees, 7 leaves, at least 150 deals per leaf), with and without a monotone constraint on discount. Table 1 reports regret on books of 20,000 deals.

Table 2. Regret of the constrained tree model at five capacities, 20,000 training deals (10 books). "Diff." is constrained minus unconstrained. "Mean rec." is the mean recommended discount at $\rho = 0$; the mean optimal discount is 8.6%.

Capacity	$\rho = -0.6$	$\rho = 0$	$\rho = +0.6$	Diff., −0.6	Diff., +0.6	Mean rec.
50 trees, 7 leaves, 150 per leaf (shipped)	22.6 (1.2)	9.0 (0.7)	1.4 (0.3)	+7.9 (0.8)	−0.5 (0.2)	5.0%
200 trees, 7 leaves, 150 per leaf	11.9 (0.8)	1.8 (0.4)	6.7 (0.5)	+4.2 (0.5)	+0.3 (0.2)	7.3%
500 trees, 7 leaves, 50 per leaf	11.0 (0.9)	1.8 (0.2)	11.0 (0.7)	+3.0 (1.3)	+1.0 (0.4)	8.2%
500 trees, 15 leaves, 50 per leaf	10.3 (1.0)	2.7 (0.3)	12.2 (0.6)	+0.1 (2.1)	+2.4 (0.3)	8.9%
500 trees, 15 leaves, 20 per leaf, rate 0.1	11.9 (0.9)	4.9 (0.7)	12.8 (0.6)	−1.5 (1.9)	+2.3 (0.3)	10.3%

Under rescue confounding the constraint raises regret by 4.3 to 11.0 points on the unrestricted sweep, as Proposition 3.5 predicts. Under whale confounding it makes little difference. The constraint does what it was introduced for: the unconstrained model produces a visibly decreasing curve for most deals under rescue confounding and the constrained model for none. Clipping recommendations to the observed range reduces the worst-case regret of the constrained model from 30.5% to 5.2% and the cost of the constraint to under a point. In this configuration it is the clip, not the constraint, that protects the decision. Results at 2,000 deals are similar (Appendix A).

6.2 Model capacity

Table 2 varies the capacity of the tree model.

Three things stand out. First, with no confounding at all the production configuration loses 9.0% of revenue, because fifty boosting rounds at a small learning rate under-read the steep initial part of the response; two hundred rounds lose 1.8%. Second, added capacity moves the exposure from one direction of confounding to the other: the small model is nearly unaffected by whale confounding and badly affected by rescue confounding, and at 500 trees the reverse holds. The small model's under-reading happens to offset the upward bias from whale confounding. Third, the cost of the constraint under rescue confounding shrinks as capacity grows and is within noise at the two largest settings, while a cost under whale confounding appears, which Proposition 3.5 does not predict and which we attribute to finite samples: the constraint removes the noisy dips that otherwise stop the model from recommending still more discount. **[TODO: Rerun with 30 books and with the clipped sweep.]**

6.3 The leniency control function

To separate the correction from the capacity of the tree model, we use a logistic win model with a cubic spline in discount, fitted without correction ("naive"), with the leniency control function ("corrected"), and with the latent quality observed ("oracle").

At 20,000 deals the corrected model is within a point of the oracle at every setting and reduces the worst regret of the naive model from 18.4% to 0.5% (Table 3, Figure 2). The naive spline model is harmed mainly by whale confounding, the opposite of the production tree model.

At 2,000 deals the corrected model again tracks the oracle, and both lose 3 to 8%. Under rescue confounding the naive model does better than either, because its bias offsets a small-sample tendency of the spline model to over-discount. The correction is doing its job; at this sample size the job is not worth doing in that direction.

Table 4 varies the number of reps. Six reps do about as well as fifty: with random assignment the number of deals, not the number of reps, limits the correction.

Table 3. Regret of a spline win model, unrestricted sweep. Columns 2 to 4: 20,000 training deals. Columns 5 to 7: 2,000.

ρ	Naive	Corrected	Oracle	Naive	Corrected	Oracle
−0.9	6.0 (0.5)	1.2 (0.4)	0.3 (0.1)	8.6 (1.5)	7.4 (1.5)	7.0 (1.5)
−0.6	2.3 (0.2)	0.5 (0.1)	0.2 (0.0)	4.2 (0.8)	8.1 (1.4)	6.9 (1.3)
−0.3	0.6 (0.1)	0.3 (0.1)	0.2 (0.0)	3.6 (0.9)	6.5 (1.2)	5.3 (1.2)
0	0.2 (0.0)	0.3 (0.0)	0.2 (0.0)	4.2 (1.1)	4.3 (1.0)	4.8 (1.1)
+0.3	4.7 (0.6)	0.3 (0.0)	0.2 (0.1)	11.4 (1.1)	4.1 (1.0)	3.8 (1.0)
+0.6	13.7 (0.7)	0.4 (0.1)	0.3 (0.1)	17.2 (0.8)	3.3 (0.8)	3.1 (0.8)
+0.9	18.4 (0.5)	0.5 (0.1)	0.3 (0.0)	18.8 (0.5)	3.3 (0.8)	3.8 (1.0)

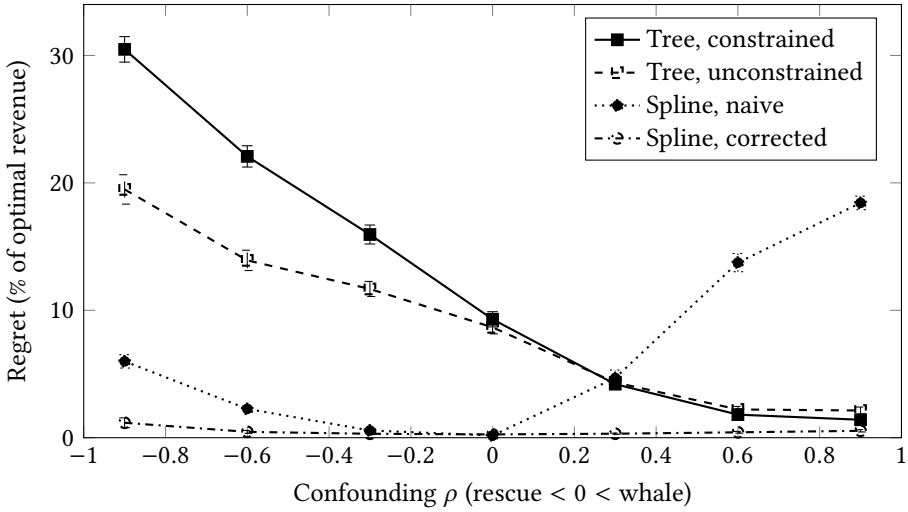


Fig. 2. Regret against the strength and direction of confounding, 20,000 training deals, unrestricted sweep. Bars are one standard error over 30 books. The tree model at production capacity is exposed to rescue confounding and the spline model to whale confounding.

Table 4. Corrected minus naive regret by number of reps. Columns 2 to 4: 20,000 training deals at $\rho = -0.6, 0, +0.6$. Columns 5 to 7: 2,000.

Reps	−0.6	0	+0.6	−0.6	0	+0.6
6	−0.8 (0.7)	0.0 (0.0)	−14.1 (0.7)	+4.8 (1.5)	+0.5 (0.8)	−9.8 (0.9)
12	+0.2 (0.9)	+0.3 (0.2)	−13.2 (0.9)	+0.7 (1.2)	+0.3 (0.7)	−10.3 (0.9)
25	−0.1 (0.7)	+0.1 (0.1)	−12.6 (0.6)	+4.0 (1.0)	+0.7 (0.6)	−10.8 (0.9)
50	−1.0 (0.4)	+0.1 (0.0)	−11.0 (0.5)	+2.2 (1.1)	0.0 (0.5)	−11.5 (0.8)

6.4 Rep skill

Table 5 adds rep skill worth 0.3 on the logit scale per standard deviation, either exactly aligned with leniency (lenient reps are weaker), partly aligned, or unrelated to it.

Table 5. Regret when reps differ in skill, 20,000 training deals. "Robust" is the minimax-regret recommendation over a sensitivity range that is calibrated to residual rep differences or fixed at ±0.3. "Segment median" recommends the segment's median historical discount.

Scenario	ρ	Naive	Corrected	Robust (calibrated)	Robust (fixed)	Segment median
valid	−0.6	2.2 (0.2)	2.9 (0.9)	2.9 (0.9)	4.0 (0.1)	6.9 (0.2)
	0	0.2 (0.1)	1.3 (0.5)	1.3 (0.5)	3.7 (0.1)	6.8 (0.2)
	+0.6	13.8 (0.9)	1.4 (0.7)	1.4 (0.7)	3.4 (0.2)	6.8 (0.1)
aligned	−0.6	9.5 (0.5)	9.5 (0.9)	9.4 (0.9)	3.2 (0.2)	6.9 (0.2)
	0	2.1 (0.2)	8.0 (0.6)	7.9 (0.6)	3.1 (0.2)	6.9 (0.2)
	+0.6	1.4 (0.5)	7.6 (0.5)	7.6 (0.5)	3.6 (0.2)	6.9 (0.1)
partial	−0.6	8.6 (0.5)	8.2 (0.9)	2.8 (0.2)	2.3 (0.2)	7.0 (0.2)
	0	1.8 (0.2)	7.0 (0.6)	2.9 (0.2)	2.3 (0.1)	7.0 (0.2)
	+0.6	1.9 (0.7)	6.6 (0.5)	3.2 (0.2)	2.7 (0.2)	7.0 (0.1)
unaligned	−0.6	2.3 (0.2)	2.8 (0.7)	4.0 (0.1)	4.0 (0.1)	6.9 (0.2)
	0	0.2 (0.0)	0.8 (0.3)	3.7 (0.1)	3.8 (0.1)	6.9 (0.2)
	+0.6	14.3 (0.9)	0.6 (0.2)	3.4 (0.2)	3.5 (0.2)	6.9 (0.1)
worst case		14.3	9.5	9.4	4.0	7.0

Table 6. The rep test: mean estimated spread of residual rep effects ($\hat{\tau}$, logit scale) and share of books flagged at the 5% level. Columns 2 and 3: 20,000 training deals. Columns 4 and 5: 2,000. The true visible spread is 0, 0, 0.29 and 0.30.

Scenario	$\hat{\tau}$	Flagged	$\hat{\tau}$	Flagged
valid	0.01	6%	0.03	7%
aligned	0.01	9%	0.03	4%
partial	0.28	100%	0.27	98%
unaligned	0.29	100%	0.29	100%

With aligned skill the corrected model loses 7.6 to 9.5%, and with no confounding or whale confounding it is far worse than the naive model. The test for residual rep differences flags these books no more often than books with a valid instrument (Table 6), as Proposition 4.1 requires. It flags skill that is not aligned with leniency in essentially every book and recovers its magnitude.

The calibrated range costs nothing when the instrument is valid and protects when some skill is visible. It cannot protect in the fully aligned case. The fixed range has the best worst case, 4.0%, at a price of about three points when nothing is wrong; it contains the true direct effect by construction in these scenarios. An alternative design, abstaining on deals whose best discount moves by more than five points across the range and falling back to the segment median, was worse than the robust recommendation it replaced by 3.2 to 4.4 points, because the fallback is itself a poor recommendation.

6.5 Non-random assignment

Finally we let assignment depend on the deal (Table 7).

Assignment by territory is flagged by the balance test in every book and is nearly harmless once segment is controlled for. Sorting on latent quality is not flagged by the balance test, since the book does not record what the sorting is based on, and it makes the corrected model lose 12 to 21% when strong deals go to lenient reps. The

Table 7. Regret under non-random assignment of reps, 20,000 training deals. "Territory": each deal goes to a rep whose home segment matches. "Sorting ±c": the correlation between a deal's latent quality and the leniency rank of its rep is c in magnitude; positive sends strong deals to lenient reps. "Balance" and "Rep test" are the share of books flagged at the 5% level.

Assignment	ρ	Naive	Corrected	Robust (fixed)	Balance	Rep test
Random	−0.6	2.2 (0.2)	2.9 (0.9)	4.0 (0.1)	3%	6%
	0	0.2 (0.1)	1.3 (0.5)	3.7 (0.1)		
	+0.6	13.8 (0.9)	1.4 (0.7)	3.4 (0.2)		
Territory	−0.6	2.3 (0.4)	4.1 (1.3)	4.6 (0.3)	100%	1%
	0	0.8 (0.4)	3.4 (1.2)	4.2 (0.3)		
	+0.6	16.5 (0.7)	4.0 (1.3)	5.0 (0.8)		
Sorting +0.3	−0.6	1.2 (0.2)	18.6 (0.8)	7.4 (0.3)	5%	29%
	0	5.3 (0.8)	15.7 (0.9)	7.1 (0.2)		
	+0.6	17.1 (0.6)	12.4 (1.0)	6.7 (0.3)		
Sorting +0.6	−0.6	1.4 (0.3)	21.2 (0.3)	13.7 (0.6)	5%	79%
	0	14.2 (0.6)	21.1 (0.3)	13.3 (0.6)		
	+0.6	18.8 (0.4)	19.5 (0.5)	11.7 (0.5)		
Sorting −0.3	−0.6	3.8 (0.3)	1.6 (0.2)	0.8 (0.1)	7%	44%
	0	0.9 (0.1)	2.8 (0.3)	1.0 (0.1)		
	+0.6	7.3 (1.3)	3.8 (0.3)	1.2 (0.2)		
Sorting −0.6	−0.6	5.2 (0.3)	5.8 (0.5)	3.0 (0.2)	7%	86%
	0	2.5 (0.2)	9.4 (0.7)	4.2 (0.2)		
	+0.6	2.2 (0.7)	15.2 (1.4)	4.5 (0.3)		

first-stage F statistic is, if anything, larger under sorting. The rep test has some power on large books, which comes from the mismatch between rank-based sorting and the additive form of the win model and should not be relied on. In nine of the twelve sorting cells the naive model has lower regret than the corrected one.

7 Guidance

[TODO: Turn into a table indexed by book size, model capacity and what is known about assignment.] The results suggest the following.

- Check capacity before confounding. A win model that under-fits the response is biased toward too little discount whatever the data.
- Do not expect a monotone constraint to fix the recommendation. Restrict recommendations to the range of discounts the cohort has actually received.
- Use a leniency instrument only where assignment follows a known rule, such as round-robin routing or territories recorded in the data. A balance test is necessary and not sufficient.
- If the corrected curve says large discounts lose deals, treat that as evidence against the instrument.
- When assignment is doubtful, prefer a recommendation that is robust over a stated range of direct effects to either trusting the correction or falling back to historical practice.
- On small books, do not correct.

8 Limitations

All outcomes are simulated. The benchmark is one family of generators with a saturating response, normally distributed latent quality, and an additive discount equation; the propositions are population statements for a single cohort. The spline win model is close to the generator's functional form. We tested one mechanism each for skill and for sorting. The paper says nothing about which strength or direction of confounding real deal books exhibit; Phillips et al. (2015) is the one estimate we know of, in auto lending.

[TODO: Conflict of interest statement. Statement on the use of AI tools.]

References

R. W. Blundell and J. L. Powell. 2004. "Endogeneity in Semiparametric Binary Response Models." *Review of Economic Studies*, 71, 3, 655–679. doi:10.1111/j.1467-937X.2004.00299.x.

C. Bockel-Rickermann, S. Verboven, T. Verdonck, and W. Verbeke. 2023. *A Causal Perspective on Loan Pricing: Investigating the Impacts of Selection Bias on Identifying Bid-Response Functions*. arXiv:2309.03730. (2023).

D. C. Chan, M. Gentzkow, and C. Yu. 2022. "Selection with Variation in Diagnostic Skill: Evidence from Radiologists." *Quarterly Journal of Economics*, 137, 2, 729–783.

D. Chetverikov and D. Wilhelm. 2017. "Nonparametric Instrumental Variable Estimation Under Monotonicity." *Econometrica*, 85, 4, 1303–1320. doi:10.3982/ECTA13639.

T. G. Conley, C. B. Hansen, and P. E. Rossi. 2012. "Plausibly Exogenous." *Review of Economics and Statistics*, 94, 1, 260–272.

W. Dobbie, J. Goldin, and C. S. Yang. 2018. "The Effects of Pretrial Detention on Conviction, Future Crime, and Employment: Evidence from Randomly Assigned Judges." *American Economic Review*, 108, 2, 201–240.

A. N. Elmachtoub and P. Grigas. 2022. "Smart "Predict, then Optimize"." *Management Science*, 68, 1, 9–26. doi:10.1287/mnsc.2020.3922.

C. Fernández-Loría and F. Provost. 2022. "Causal Decision Making and Causal Effect Estimation Are Not the Same…and Why It Matters." *INFORMS Journal on Data Science*, 1, 1, 4–16. doi:10.1287/ijds.2021.0006.

B. Frandsen, L. Lefgren, and E. Leslie. 2023. "Judging Judge Fixed Effects." *American Economic Review*, 113, 1, 253–277. doi:10.1257/aer.20201860.

M. Ghoummaid and U. Shalit. 2024. "When to Act and When to Ask: Policy Learning With Deferral Under Hidden Confounding." In: *Advances in Neural Information Processing Systems*.

N. Kallus and A. Zhou. 2021. "Minimax-Optimal Policy Learning Under Unobserved Confounding." *Management Science*, 67, 5, 2870–2890.

Y. Karlinsky-Shichor and O. Netzer. 2024. "Automating the B2B Salesperson Pricing Decisions: A Human-Machine Hybrid Approach." *Marketing Science*, 43, 1, 138–157. doi:10.1287/mksc.2023.1449.

J. R. Kling. 2006. "Incarceration Length, Employment, and Earnings." *American Economic Review*, 96, 3, 863–876.

J. Mandi, J. Kotary, S. Berden, M. Mulamba, V. Bucarey, T. Guns, and F. Fioretto. 2024. "Decision-Focused Learning: Foundations, State of the Art, Benchmark and Future Opportunities." *Journal of Artificial Intelligence Research*, 80, 1623–1701. doi:10.1613/jair.1.15320.

A. Petrin and K. Train. 2010. "A Control Function Approach to Endogeneity in Consumer Choice Models." *Journal of Marketing Research*, 47, 1, 3–13.

R. Phillips, A. S. Şimşek, and G. van Ryzin. 2015. "The Effectiveness of Field Price Discretion: Empirical Evidence from Auto Lending." *Management Science*, 61, 8, 1741–1759. doi:10.1287/mnsc.2014.2084.

A Additional Tables

Table 8. As Table 1, with 2,000 training deals.

ρ	Constrained	Unconstrained	Difference	Constrained (clip)	Difference (clip)
−0.9	34.8 (1.8)	24.7 (2.5)	+10.1 (1.8)	5.3 (0.3)	+1.3 (0.2)
−0.6	27.7 (1.8)	19.9 (2.1)	+7.8 (1.3)	4.2 (0.3)	+0.7 (0.2)
−0.3	21.5 (1.8)	16.4 (2.0)	+5.2 (0.9)	3.1 (0.3)	+0.4 (0.1)
0	14.0 (1.4)	11.9 (1.5)	+2.2 (0.7)	2.5 (0.3)	+0.1 (0.1)
+0.3	9.8 (1.1)	8.8 (1.1)	+0.9 (0.4)	2.1 (0.2)	+0.1 (0.1)
+0.6	7.7 (0.9)	7.5 (0.9)	+0.2 (0.2)	2.1 (0.2)	0.0 (0.1)
+0.9	5.4 (0.7)	5.8 (0.7)	−0.4 (0.2)	2.7 (0.3)	0.0 (0.1)

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(5) All data sets drawn from the existing literature (potentially including authors' own previously published work) are publicly available. [NA]
(6) All new data sets and data sets that are not publicly available are described in detail, including relevant statistics, the data collection process and annotation process if relevant. [partially]
(7) All methods used for preprocessing, augmenting, batching or splitting data sets (e.g., in the context of hold-out or cross-validation) are described in detail. [partially]

Explanations on any of the answers above (optional):

These are draft answers for the author to confirm. Proofs are given in full but have not yet been checked independently. Code, raw results and the benchmark generator exist in the author's repository; the public location and licence are not yet chosen. Randomness: every book is generated from a fixed integer seed and the scripts list the seeds, but the article does not yet describe this. The execution environment is not yet described. Hyperparameters: the tree model uses a production configuration and a five-point capacity sweep is reported; no

other tuning was done. Statistics: paired differences with standard errors over 30 books are reported; no formal hypothesis tests. The only data are generated by the benchmark; no data sets from the literature are used.